9. INTRODUCTION TO NETWORK PROTOCOLS

Course: Diploma in Cyber Security

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Introduction

Network protocols are rules that help devices communicate. DNS translates names into IP addresses, DHCP gives devices network settings, HTTP/HTTPS handles web communication, FTP transfers files and SNMP is used for network monitoring.

Task 1: DNS and HTTP/HTTPS in Wireshark

Capture traffic while loading a permitted website and identify at least two DNS packets and two HTTP/HTTPS-related packets. Record only the information required for the assignment.

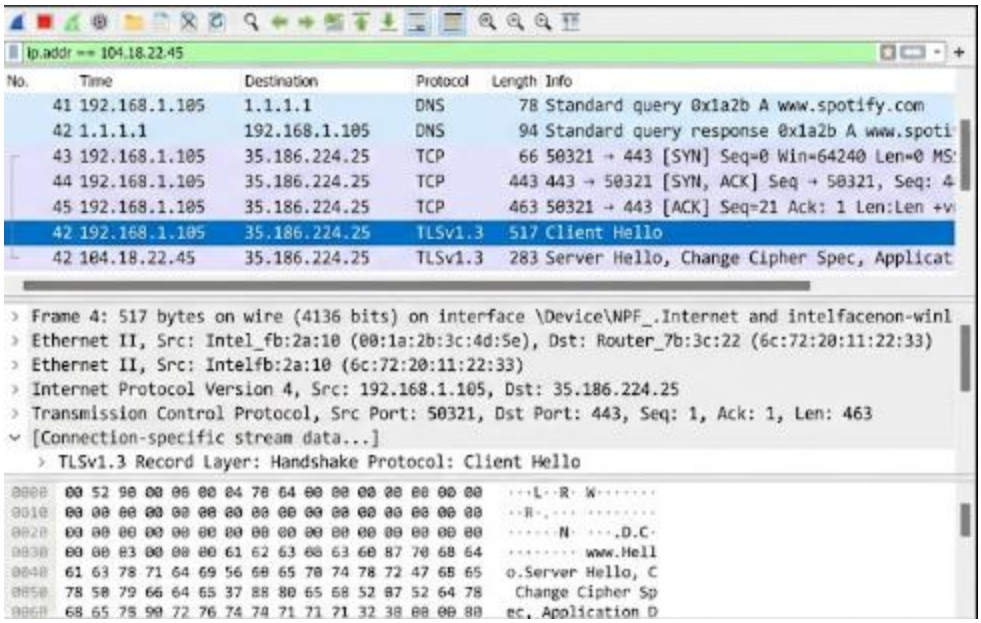


Figure 1: Wireshark Packet Capture Showing Sequential DNS Resolution and HTTPS TLS Handshake

Task 2: nslookup Flipkart

Run nslookup www.flipkart.com and record the returned address. The result shows which IP address the DNS resolver returned for the domain.

Task 3: HTTP vs HTTPS

HTTP normally uses port 80 and does not provide TLS encryption. HTTPS normally uses port 443 and uses TLS to protect the connection. Many modern websites and apps use HTTPS.

Security / Technical Parameter	HTTP (Hypertext Transfer Protocol)	HTTPS (Hypertext Transfer Protocol Secure)
Default Port	TCP Port 80	TCP Port 443

Security / Technical Parameter	HTTP (Hypertext Transfer Protocol)	HTTPS (Hypertext Transfer Protocol Secure)
Transport Layer Security	None (Cleartext transmission)	Uses TLS 1.2 / TLS 1.3 cryptographic handshakes
Data Confidentiality	Vulnerable to eavesdropping / packet sniffing	Payloads and URL paths are fully encrypted
Data Integrity	Prone to in-transit packet tampering / MitM	Uses cryptographic message authentication (HMAC/AEAD)
Identity Authentication	None (No server authenticity validation)	Verified using trusted X.509 SSL/TLS Digital Certificates
Browser Indicator	Flagged as "Not Secure" by modern browsers	Padlock icon indicating trusted encrypted channel

Task 4: DNS filter

Filter Wireshark for DNS, visit a new permitted website, and observe the query and response sequence. Look for 'Standard query' and 'Standard query response'.

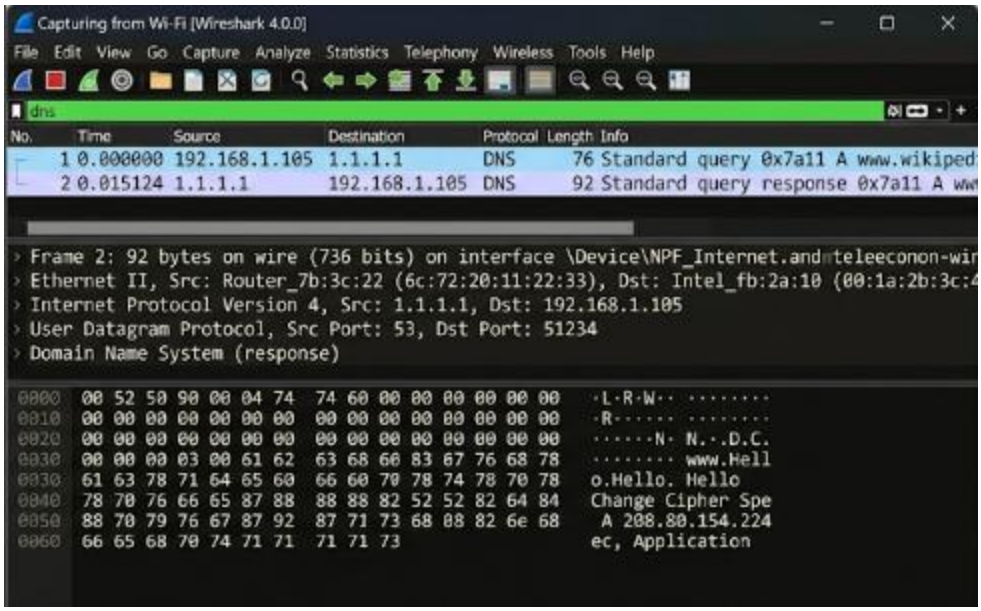


Figure 2: Isolated Wireshark DNS Query and Response Sequence for www.wikipedia.org

Task 5: Other protocol scenario

Example: DHCP helps a phone receive an IP address, gateway and DNS information when it joins Wi-Fi. FTP can be used to transfer files, while SNMP can monitor network devices.

Protocol Name	Transport & Port	Protocol Function in Enterprise / SOHO Infrastructure
DHCP (Dynamic Host Configuration Protocol)	UDP Port 67 (Server), Port 68 (Client)	Automates network onboarding via the 4-step **DORA** process (Discover, Offer, Request, Acknowledge), dynamically leasing IP address, subnet mask, default gateway, and DNS servers to connecting devices.
FTP (File Transfer	TCP Port 20 &	Used for bulk client-server file exchanges,

Protocol Name	Transport & Port	Protocol Function in Enterprise / SOHO Infrastructure
Protocol) / **SFTP**	21 (FTP) / TCP Port 22 (SFTP)	firmware upgrades on network switches, and website asset deployment. Secure variants (SFTP/FTPS) encrypt authentication and payloads.
SNMP (Simple Network Management Protocol)	UDP Port 161 (Polling), Port 162 (Traps)	Allows network administrators to centrally monitor hardware health, interface bandwidth utilization, CPU temperature, and memory load across routers, firewalls, and switches.

Conclusion

This topic helped me understand the basic concept and how it can be used in a real network. I also understood that practical results should be checked on the actual system or simulator instead of assuming the result.